

“A segment where we see significant growth are solid-state drives”

Be it in mobile, removable or cloud, 40 zettabytes of memory will be the requirement in future, claims Dr Siva Sivaram, senior vice president, Memory Technology, SanDisk Corporation, as he talks to Abhishek Mutha of EFY. 2D NAND would take a back seat in the future due to its limitations and 3D NAND technology would be in the driver's seat by 2020. In this interview, he talks about the significance and widespread use of Flash based storage solutions, upcoming new technologies and impact of solid-state drives



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Q. What is your take on the memory industry, in terms of requirement today?

In the memory world, the demand for memory is growing enormously. The average density in a mobile product is expected to grow to about 26 gigabytes. By 2020, every man, woman and child in the world is going to need about an additional 8 to 10 gigabytes of memory. 40 zettabytes (equivalent to 40 billion 1TB drives) of memory will be the requirement in future.

Q. How is Flash storage technology growing to keep up with this requirement?

The growth is in odds, trying to satisfy that number in all its form factors—removable, mobile/embedded, and cloud and enterprise. So, with respect to high densities, this decade we are probably in good shape with 3D NAND memories. However, in the later part of this decade, we will start to see new technologies coming in to make even higher density memories.

Q. Are there any upcoming new memory technologies generating attention as successors to NAND Flash that could lead to boosts in performance and endurance far beyond the current state of the art?

Today, 3-bit-per-cell (X3) technology, advanced architectures and other Flash memory innovations are enabling new types of products. Tomorrow, our research into 3D memory and other types of memory solutions will drive the future of the storage industry and deliver an exciting new range of consumer and business devices built on SanDisk innovations.

Q. What is the status with 3D NAND memory?

As a pioneer and leader within the industry, SanDisk is driving research into 3D NAND. This adds a third

dimension to the traditional X and Y, thereby giving 3D NAND storage technologies the ability to continue capacity and cost scaling that have enabled many markets to be created over the past decades. This technology enables devices to outperform the previous generation of memory devices, which were based on the 20nm planar NAND Flash technology. 3D NAND technology allows us to tackle problems that are usually faced by planar 2D NAND technology—especially with the increase in memory density.

Q. What are the biggest advantages users can get from Flash storage?

Flash storage offers a number of advantages to enterprise companies. These include the ability to access critical data more quickly while consuming less power and less real estate. Expanding the use of Flash in data centres is a key strategic focus for SanDisk, and we recently acquired SMART storage systems to bolster our existing enterprise offerings. Today, SanDisk's enterprise SSD solutions are qualified at six of the top seven storage OEMs, and we see this market offering strong growth opportunities in the years to come.

Q. Any particular segment where you are seeing significant growth with respect to Flash?

A segment where we see significant growth are solid-state drives—SSDs. Flash-based SSDs have already had a major impact on the laptop/notebook market. And, as enterprises and data centres increasingly look for ways to reduce costs, while improving their abilities to provide fast access to critical information, SSDs will play an expanding role in meeting these needs, thereby giving companies such as SanDisk exciting growth opportunities in the enterprise/data centre market. ●